## 11.6 Theoretical Distributions: Some Special Discrete Distributions

In the previous section we have dealt with various general probability distributions with mean and variance. We shall now learn some discrete probability distributions of special importance.

In this section we learn the following discrete distributions.

(i) The One point distribution
(ii) The Two point distribution
(iii) The Bernoulli distribution
(iv) The Binomial distribution.

### 11.6.1 The One point distribution

The random variable $X$ has a one point distribution if there exists a point $x_0$ such that, the probability mass function $f(x)$ is defined as $f(x) = P(X = x_0) = 1$.

That is the probability mass is concentrated at one point.

The cumulative distribution function is

$$F(x) = \begin{cases} 0 & -\infty < x < x_0 \\ 1 & x_0 \le x < \infty \end{cases}$$

**Mean :**

$$E(X) = \sum_x x\, f(x) = x_0 \times 1 = x_0$$

**Variance :**

$$V(X) = E(X^2) - \left(E(X)\right)^2 = \sum_x x^2 f(x) - \left(x_0\right)^2 = {x_0}^2 - {x_0}^2 = 0$$

Therefore the mean and the variance are respectively $x_0$ and $0$.

### 11.6.2 The Two point distribution

**(a) Unsymmetrical Case:** The random variable $X$ has a two point distribution if there exists two values $x_1$ and $x_2$, such that

$$f(x) = \begin{cases} p & \text{for } x = x_1 \\ 1-p & \text{for } x = x_2 \end{cases} \qquad \text{where } 0 < p < 1.$$

The cumulative distribution function is

$$F(x) = \begin{cases} 0 & \text{if } x < x_1 \\ p & \text{if } x_1 \le x < x_2 \\ 1 & \text{if } x \ge x_2 \end{cases}$$

**Mean :**

$$E(X) = \sum_x x\, f(x) = x_1 \times p + x_2 \times (1-p) = px_1 + qx_2 \text{ where } q = 1-p.$$

**Variance :**

$$V(X) = E(X^2) - \left(E(X)\right)^2 = \sum_x x^2 f(x) - \left(px_1 + qx_2\right)^2$$

$$= \left({x_1}^2 p + {x_2}^2 q\right) - \left(px_1 + qx_2\right)^2 = pq\left(x_2 - x_1\right)^2$$

The mean and the variance are respectively $px_1 + qx_2$ and $pq\left(x_2 - x_1\right)^2$



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